J. theor. Biol. (2000) **204**, 15–20
doi:10.1006/jtbi.2000.1082, available online at http://www.idealibrary.com on IDEAL®

On the Classes of Aminoacyl-tRNA Synthetases and the Error Minimization in the Genetic Code

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(*Received on* 12 *May* 1999, *Accepted in revised form on* 20 *December* 1999)

As a consequence of the existence of two classes of aminoacyl-tRNA synthetases (aaRSs), we defined two types of mutations: g (mutations that do not change the class of the involved amino acids) and u (those which change the class). We have found that the mean chemical distance resulting from g mutations is smaller than that corresponding to u mutations, indicating that g mutations are responsible for most of the known minimization of the genetic code. This supports models for the origin and evolution of the code, in which new amino acids were added after duplications or modification of existing aaRSs.

Introduction

Since the genetic code was deciphered in the 1960s, molecular biologists noticed that similar codons often specify similar amino acids. In this way, the effects of point mutation or mistranslation are minimized, because mutated codons are either synonymous with or code for an amino acid with chemical properties very similar to those of the original codon.

Many authors have tried to quantify the strength of this association between codon position and amino acid properties. Salemme *et al.* (1977), and Wong (1980) have questioned whether most non-synonymous single-base changes do, in fact, substitute similar amino acids. Di Giulio (1989a) estimated that the natural code has achieved 68% minimization of polarity distances. More recently, Haig & Hurst (1991) showed that the minimization is even higher than that obtained by previous studies, and their results have been extended by Freeland & Hurst (1998).

None of these studies, however, takes into account the fact that the aminoacyl-tRNA synthetases (aaRS), whose function is to charge the tRNAs with their cognate amino acids, are descendants of two ancestral enzymes (Eriani *et al.*, 1990, 1995; Cusack *et al.*, 1990, 1991), hence can be divided into two classes. Wetzel (1978) called the attention to the fact that these enzymes could have played a role in code organization, and in a recent paper (Wetzel, 1995) he pointed out that there is a correlation between the code organization and the division of the synthetases into two classes. All codons of the type XCX code for amino acids of Class II, and those of the type XUX usually code for amino acids of Class I, except UUU and UUC, which code for Phe. The other columns of the code do not present any preference for amino acids of a given class (Table 1).

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0022–5193/00/090015 + 06 $35.00/0

An interesting feature of the classes of aaRS is that they divide the amino acids into two groups of ten residues, each forming a complete set with respect to the usually considered chemical characters, hydrophobicity, hydrophilicity, acid and basic properties (Table 2). Also, the amino acids corresponding to Class II synthetases tend to be smaller than those of Class I.

Based on the two different classes of aaRS, we can define two types of mutations: those in which the initial and final amino acids belong to different classes, which we denote u mutations, and mutations in which the amino acids belong to the same class, which we call g mutations (excluding the synonym mutations, in which the initial and final amino acids are the same). The g/u notation comes from the German gerade (even) and ungerade (odd), used in Group Theory.

TABLE 1

The genetic code, showing the codons cognate to each class of aaRS. All codons in the second column code for amino acids of Class II, while those in the first column (except UUU and UUC) code for amino acids of Class I

1ª. base	Codon 2ª. base U	C	A	G	3ª. base
U	*Phe II*	*Ser II*	Tyr I	Cys I	U
	Phe II	*Ser II*	Tyr I	Cys I	C
	Leu I	*Ser II*	Stop	Stop	A
	Leu I	*Ser II*	Stop	Trp I	G
C	Leu I	*Pro II*	*His II*	Arg I	U
	Leu I	*Pro II*	*His II*	Arg I	C
	Leu I	*Pro II*	Gln I	Arg I	A
	Leu I	*Pro II*	Gln I	Arg I	G
A	Ile I	*Thr II*	*Asn II*	*Ser II*	U
	Ile I	*Thr II*	*Asn II*	*Ser II*	C
	Ile I	*Thr II*	*Lys II*	Arg I	A
	Met I	*Thr II*	*Lys II*	Arg I	G
G	Val I	*Ala II*	*Asp II*	*Gly II*	U
	Val I	*Ala II*	*Asp II*	*Gly II*	C
	Val I	*Ala II*	Glu I	*Gly II*	A
	Val I	*Ala II*	Glu I	*Gly II*	G

In this paper, we show that the division of the 20 amino acids according to the classes of aaRS is such that the mean chemical distance between all pairs of amino acids belonging to the same class is smaller than that obtained for 99% of all other possible partitions of the amino acids in two groups. We also show that mutations involving codons specifying amino acids of the same class (excluding synonym mutations) lead to smaller chemical distances than those that do change the class. In other words, the largest contribution to the minimization of the code as a whole is due to g mutations. We interpret this as an indication that the existence of two different classes of aminoacyl-tRNA synthetases played an important role in the organization of the code.

Many scales have been proposed in which it is possible to calculate the distance between two amino acids. Shuichi *et al.* (1999) present a list of 434 indices and 66 matrices, which could be used for this purpose. Many of these indices, however, refer to properties of amino acids residues within proteins, and not to the free amino acids.

Di Giulio (1989b) showed that codon assignment can be correlated to amino acid polarity and volume, and that the strongest correlation occurs for the index proposed by Miyata *et al.* (1979), which is based on the differences of polarity and volume between two given amino acids. In what follows, we use the Miyata index as

TABLE 2

The classes of aaRS divide the amino acids into two chemically complete sets

Chemical character	Class I	Class II
Hydrophobicity	Ile, Leu, Met, Trp, Val	Ala, Phe, Pro
Hydrophilicity	Cys, Gln, Tyr	Asn, Gly, Ser, Thr
Acidic	Glu	Asp
Basic	Arg	His, Lys

a measure of the distance between different amino acids.

Methods

The distance defined by Miyata *et al.* (1979) is given by

$$d_{ij} = \sqrt{\left(\frac{\Delta p_{ij}}{\sigma_p}\right)^2 + \left(\frac{\Delta v_{ij}}{\sigma_v}\right)^2} \tag{1}$$

where Δp_{ij} and Δv_{ij} stand for the absolute values of polarity and volume differences between amino acids i and j, respectively (i.e. $\Delta p_{ij} = |p_i - p_j|$ and $\Delta v_{ij} = |v_i - v_j|$), and σ_p and σ_v are the standard deviations of Δp_{ij} and Δv_{ij}, respectively, used for scaling the d_{ij} values.

There are 576 possible point mutations in the genetic code (each position in each of the 64 codons can change to another three bases; thus, $3 \times 3 \times 64 = 576$). Of these, 50 mutations involve Stop codons and, as we cannot define a distance value in these cases, they will not be considered; 134 are synonym mutations and have a distance value of 0; to avoid a bias in favor of our arguments they will not be considered either. The remaining 392 are missense mutations, and can be divided into 196 u mutations, and 196 g mutations.

The Miyata distances between all possible pairs of the 20 amino acids were sorted according to the classes of the amino acids. This gives a set of 180 pairs involving amino acids of the same class and 200 pairs involving amino acids of different classes. It is important to note that not all of these 180 or 200 pairs are interconvertible by point mutations in the genetic code, but they will be used to separate the effect of code organization from that of the simple division of the amino acids into two classes in the minimization. We will label these pairs as g' and u' pairs.

Results and Discussion

The histograms in Fig. 1 show how Miyata distances are distributed for (a) all possible pairs of amino acids, (b) all pairs of type g', (c) all pairs of type u', (d) all pairs of amino acids which are interconvertible by a point mutation, (e) all g mutations, and (f) all u mutations.

The values of the mean, standard deviation, and skewness of the six distributions are shown

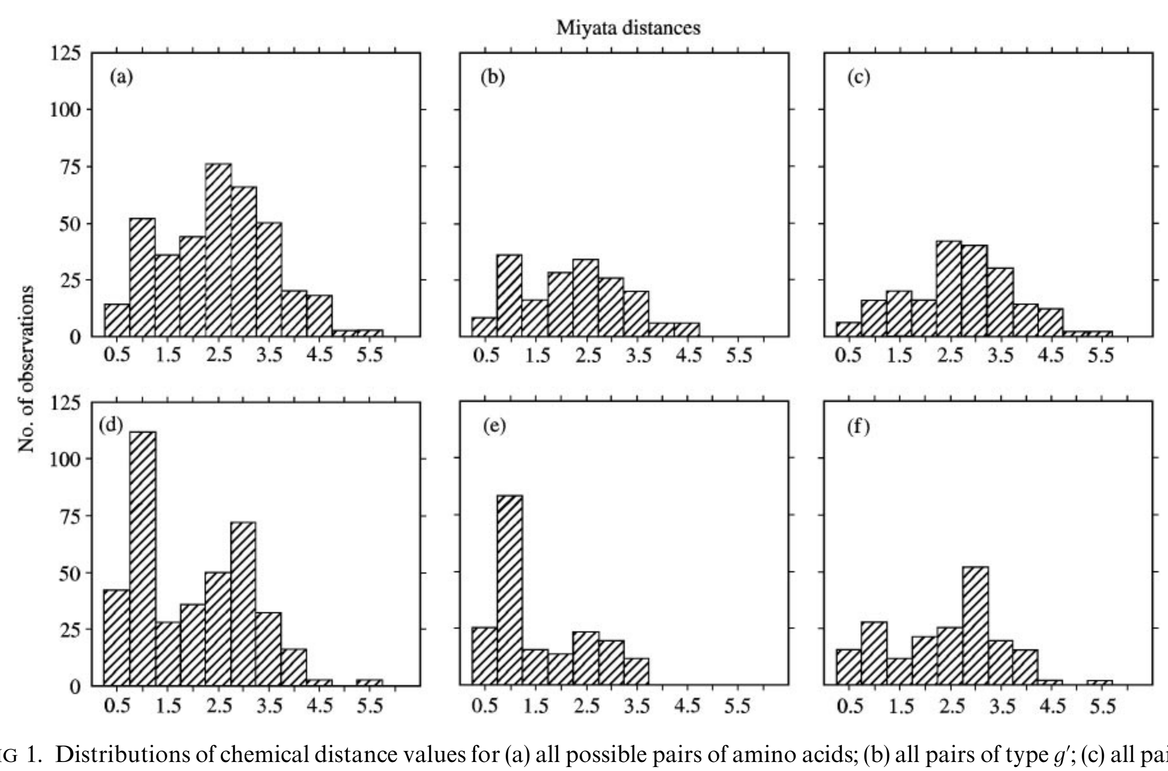

FIG 1. Distributions of chemical distance values for (a) all possible pairs of amino acids; (b) all pairs of type g'; (c) all pairs of type u'; (d) all pairs of amino acids which are interconvertible by a point mutation; (e) all g mutations; (f) all u mutations.

TABLE 3

Mean, standard deviation, and skewness values for the chemical distance distributions in Fig. 1

	Mean	S.D.	Skewness
All pairs of AA	2.26	1.05	0.069
g' pairs	2.01	1.00	0.114
u' pairs	2.49	1.04	− 0.004
All point mutations	1.77	1.10	0.296
g mutations	1.36	0.93	0.633
u mutations	2.18	1.09	− 0.100

in Table 3. The calculation of confidence intervals is not necessary here, because the distributions of distances in Fig. 1 refer not to samples, but to the populations themselves, which can be compared directly.

Although the six distributions overlap to a large extent, their statistical parameters differ in several points which are worth noting. The mean distance between amino acid pairs inside each class ($\overline{dg'} = 2.01$) is smaller than the mean distance between amino acids of different classes ($\overline{du'} = 2.49$). However, if we look only at the distances between amino acids that are interconvertible by a point mutation, this distance difference increases sharply, the mean of the distance inside each class, $\overline{dg}$, being 1.36, whereas that of amino acids of different classes, $\overline{du}$ being 2.18.

The average distance between an amino acid pair decreases by 19% as one goes from the set containing amino acids of different classes to the one restricted to pairs from the same class (that is, from $\overline{du'} = 2.49$ to $\overline{dg'} = 2.01$). When these sets are paired down to amino acids pairs that are interconvertible by a point mutation, the average distance falls from $\overline{du} = 2.18$ to $\overline{dg} = 1.36$, a 37% reduction. The differences between the comparisons $g' \rightarrow g$ and $u' \rightarrow u$ present a similar pattern: 32 and 12%, respectively.

The skewness, which measures the asymmetry of the distribution, also varies according to the classification scheme. The distributions of amino acids pairs and mutations from the same class (g' and g) are skewed to the right, that is, their skewness values are positive. The opposite is true for the distributions of pairs and mutations from different classes, which are left-skewed and have larger mean values. The skewness and mean values of the distributions containing all pairs and all point mutations are somewhere in between, since these distributions contain all u' and g' pairs and all u and g mutations, respectively.

If we consider every possible division of the 20 amino acids into two classes we verify that, of the $2^{20} = 1\,048\,576$ possible divisions, only 10 066 (0.96%) give a mean distance value for g' pairs smaller than that for the actual division, and only 3402 (0.32%) give a mean distance value for g mutations smaller than that of the actual value. These results mean that the probability that the observed minimization of the g' pairs and g mutations could have arisen purely by chance in any two partition of the amino acids is very small.

An examination of the amino acid groupings in which the mean distance value for g' pairs or g mutations are smaller than that of the actual division between Classes I and II, shows that they share many similarities. To study this, we have calculated the number of times each pair of amino acids belong to the same category in all these different groupings. We assumed that when an amino acid pair has a frequency higher than 80% (that is, the amino acids in this pair belong to the same group in at least 80% of these divisions) or smaller than 20%, its amino acids are normally present in the same or in opposite groups, respectively.

Using this procedure we could assign groups to 13 amino acids, namely: Ala, Asn, Asp, Gly, Pro, Ser, and Thr, which are in the same group, while Arg, Ile, Leu, Phe, Trp, and Tyr are in the other. This result agrees with the actual division of the amino acids into two classes, with the exception of Phe, which is not in the expected group. Since the other amino acids are neither specially frequent nor rare, we were not able to assign them to a group with this procedure.

Examining Classes I and II separately, we calculate that the mean distance for the g' pairs for Class I (that is, only the pairs of amino acids in which both belong to Class I) is 2.009, and that for Class II is 2.012, indicating that the distances between amino acids inside each class are very similar. However, when we look at those pairs of amino acids that are interconvertible by a point mutation, we observe a value of 1.575 for

g mutations of Class I, and a smaller value of 1.201 for g mutations of Class II.

This agrees with the proposals that Class II aaRSs are older than Class I aaRSs (Hartman, 1995; Ferreira & Cavalcanti, 1997; Trifonov & Bettecken, 1997). If the amino acids of Class II were incorporated earlier in the code than those of Class I, then we should expect them to be better positioned in the code than those of Class I, which had to displace codons of Class II amino acids.

Haig & Hurst (1991) showed that, considering synonym mutations, the genetic code is minimized with respect to point mutations in the first and third codon positions, while mutations in the second position usually do not give rise to similar amino acids. To explain this they note that the second position usually defines the hydrophobicity of the coded amino acid, and also that there is no synonym mutation amongst the point mutations involving the second base (while there are four and 126 in the first and third bases respectively). For the third position we believe that the minimization could be due to synonym mutations, but the strong difference between the minimization of the first and second bases deserves further analysis.

When we exclude the synonyms mutations we obtain a value of 1.41 for the mean distance of amino acids resulting from point mutations in the first base of codons, and a value of 2.24 for mutations in the second base. An analysis of these mutations shows that from the 166 missense mutations in the first position, 108 are g mutations and only 58 are u mutations; in the second position there 176 missense mutations, divided into 70 g and 106 u mutations: the higher minimization in the first base is due to the fact that g mutations have smaller distance values than u mutations.

To see if our results are not restricted to Miyata's distances, we also performed calculations using three other indices. The results, shown in Table 4, are similar and lead to the same overall conclusions.

The values in the second and fifth lines of Table 4 are larger than those in the third and fourth lines respectively, which indicate that the results obtained for Miyata's index should be valid for the others. Another interesting feature is that the results for the molecular volume index are the most marked, with the value of the mean distance increasing when we change from u' pairs to u mutations (hence the negative sign). This apparently abnormal behavior of the molecular volume index can be assigned to the large difference in the size of the amino acids between the two classes; the amino acids of Class II are usually smaller than those of Class I (Eriani *et al.*, 1995; Hartman *et al.*, 1995). The mean molecular weight for the Class I amino acids is 138, while that for Class II is 113.

Conclusion

The results presented in this paper support models for the origin and evolution of the genetic code in which the code evolved through incorporation of new amino acids following duplications of the tRNAs and tRNA synthetases, for example Crick's frozen accident hypothesis (Crick, 1968).

TABLE 4

Percent reduction in the average distance for different types of pairs and mutations. For the molecular volume, the change from all u′ pairs to all u mutations leads to a slight mean distance increase

	Miyata distance*	Grantham distance†	Molecular volume†	Polar requirement‡
All pairs → all mutations	21.7%	18.5%	13.2%	26.2%
g' pairs → g mutations	32.3%	24.1%	31.8%	27.8%
u' pairs → u mutations	12.4%	13.2%	− 2%	24.6%
u' pairs → g' pairs	19.3%	12.6%	23.6%	12.3%
u mutations → g mutations	37.6%	23.6%	48.9%	15.9%

* Miyata *et al.* (1979). † Grantham (1974). ‡ Woese *et al.* (1973).

According to this model early versions of the code specified fewer amino acids than the modern code, but most codons specified amino acids. Additional amino acids were substituted gradually, when they were able to confer selective advantages, until eventually the code became frozen in its present form. Similar amino acids tended to have similar codons because (1) this diminished the deleterious effects of the initial substitution, and (2) the new tRNA and aaRS might have been duplications or modifications of the old tRNA and synthetase, in which case the new amino acid would most likely be structurally similar to the old amino acid (Crick, 1968; Haig & Hurst, 1991).

The first factor acts in the code as a whole and predicts that all mutations should be minimized, with no difference in the minimization of u and g mutations. However, the second factor is a structural constraint: following a duplication or modification of a synthetase the new amino acid should be structurally similar to the old one, considering the similarity between the enzymes. Enzymes belonging to Class I probably appeared after duplications of an ancestor of Class I, and the same holds for the enzymes of Class II. As the new tRNA would also be a modification or duplication of the old tRNA, one should also expect the codon to be one of the neighbors of the old one. This factor predicts g mutations to be more conservative, because the amino acids belonging to a given class could only be incorporated following a duplication or modification of an existing tRNA and synthetase of the same class, and as such would be similar to and share a near codon with the old one.

This work was supported in part by CNPq (Brazilian Government Agency).

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